# Hybrid versus standard tensor trains under time evolution: benchmark of the explicit solver with SVD truncation

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The hybrid tensor-train encoding is benchmarked against a standard quantized tensor train for the time evolution of a passive scalar, at four resolutions from $256^2$ to $2048^2$. Both encodings are advanced by the same explicit solver: exact MPO–MPS contraction followed by a two-sweep SVD truncation. At a fixed target accuracy the hybrid encoding reaches that accuracy with a factor 1.8–2.5 fewer parameters, and the factor is constant across the resolutions tested. The head size that achieves it holds half the bits of each axis, $N \simeq L$, corresponding to a $2^{L/2} \times 2^{L/2}$ coarse grid represented densely and the finer scales carried by the tail. The error made in evolving tracks the error of merely representing the same field within a factor 1.5–25 for every encoding, resolution and budget measured.

## I. SETUP

The benchmark problem is a decaying passive scalar on a periodic square,

$$\partial_t \theta + \mathbf{u} \cdot \nabla \theta = \kappa \nabla^2 \theta, \qquad \nabla \cdot \mathbf{u} = 0, \tag{1}$$

with no forcing, so that the variance budget is

$$\frac{d}{dt}\frac{\langle \theta^2 \rangle}{2} = -\kappa \langle |\nabla \theta|^2 \rangle. \tag{2}$$

The velocity is the steady cellular flow obtained from the streamfunction $\psi = \sin x \, \sin y$, so $\mathbf{u} = (\sin x \cos y, \, -\cos x \sin y)$. It is constructed by applying the solver's own derivative MPOs to the encoded $\psi$, which makes it divergence-free to $7 \times 10^{-13}$ at the discrete level and gives it bond dimension 4. The initial condition is a random-phase superposition of all modes with $|\mathbf{k}| \leq 2$, normalised to zero mean and unit variance.

Space is discretised with second-order centred stencils for both the transport and the diffusion terms, and the transport term is written in skew form, $\frac{1}{2}$(advective + conservative), which inherits the antisymmetry of the continuous operator and conserves $\langle \theta^2 \rangle$ to round-off in the absence of truncation. Time integration uses SSPRK3 with a step satisfying both the advective CFL condition and the two-dimensional diffusive limit $\Delta t \leq \Delta x^2 / 4\kappa$.

Resolution is varied at fixed $\ell_B / \Delta x = 2.88$ by scaling $\kappa \propto \Delta x^2$, anchored at $\kappa = 5 \times 10^{-3}$ for $L = 8$, so that each resolution resolves the same range of scales. All runs integrate to $T = 8$, which is past the time at which the truncation error saturates.

Both encodings are advanced by identical code. The standard tensor train is the case $N = 0$ of the hybrid encoder, verified to reproduce `tt.vector_to_mps(vec,2,2L)` bond for bond. Compression policy, contraction order, stencils, velocity, time step and dense reference are shared, so the head size is the only variable between the two.

## II. HEAD SIZE: WHAT $N$ CONTROLS

The field is encoded on $2L$ binary legs in Morton order, even legs carrying the row index and odd legs the column index, with level 0 the most significant bit. The head fuses the first $N$ legs into a single dense tensor of physical dimension $2^N$; the remaining $2L - N$ legs are ordinary binary cores.

$N$ therefore selects how many of the *most significant* bits of each axis are held densely. The head holds $N/2$ bits per axis, which is a $2^{N/2} \times 2^{N/2}$ grid of coarse blocks, each block spanning $2^{L-N/2}$ cells on a side; the structure inside a block is carried by the tail. Table I gives the correspondence.

The useful range is $N \simeq L$, i.e. half the bits of each axis. The three sizes $N \in \{L-2, \, L, \, L+2\}$ bracket the optimum at every resolution measured, and the results below are reported on that bracket.

Two distinct normalisations of the head cost are worth separating, since they behave differently as $L$ grows (Fig. 1). The head occupies $2^N \chi_0$ parameters. Relative to the compressed state this is a large and increasing share: at $L = 10$, $N = 10$, $\chi = 32$ it is 74% of the stored parameters. Relative to the dense field it is $2^N \chi_0 / 4^L$, which for $N \simeq L$ is $\chi_0 / 2^L$ and therefore decreases with resolution: the same configuration stores a head worth 3.1% of the dense field, and the complete state worth 4.2%.

| $N$ | bits/axis | blocks | cells per block | | |
|---|---|---|---|---|---|
| | $N/2$ | $2^{N/2} \times 2^{N/2}$ | $L = 9$ | $L = 10$ | $L = 11$ |
| 4 | 2 | $4 \times 4$ | 128 | 256 | 512 |
| 6 | 3 | $8 \times 8$ | 64 | 128 | 256 |
| 8 | 4 | $16 \times 16$ | 32 | 64 | 128 |
| 10 | 5 | $32 \times 32$ | 16 | 32 | 64 |
| 12 | 6 | $64 \times 64$ | 8 | 16 | 32 |

TABLE I. Head size in terms of the grid it resolves densely. $N = 10$ at $L = 10$ holds 5 of the 10 bits of each axis, i.e. a $32 \times 32$ coarse grid, with the remaining $32 \times 32$ cells inside each block carried by the tail.

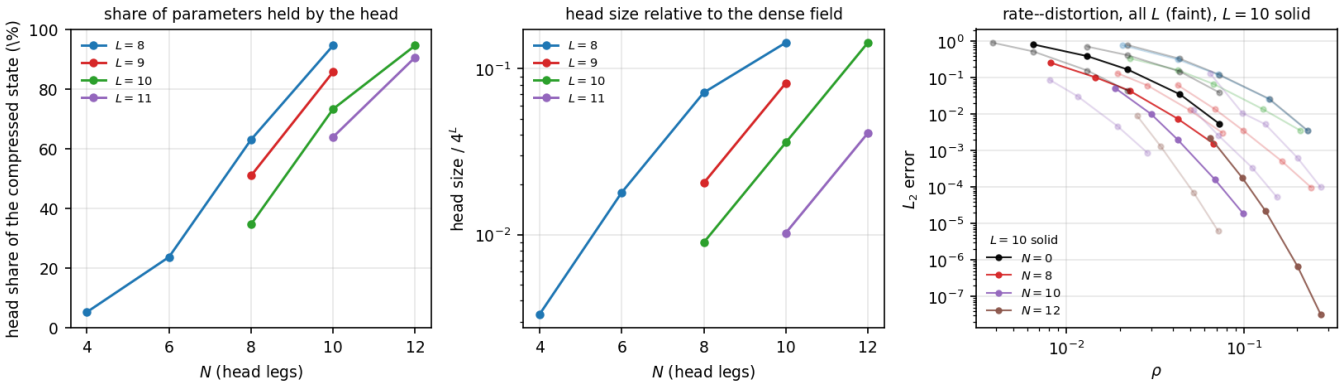



FIG. 1. Head cost under both normalisations, and the rate–distortion curves from which the optimum is read. Left: share of the compressed state held by the head. Centre: head size as a fraction of the dense field, which falls with $L$ at fixed $N - L$. Right: $L_2$ error against compression ratio $\rho$ for every run; $L = 10$ drawn solid.

## III. RATE–DISTORTION

Bond caps are integers, so the two encodings do not land on identical parameter counts. Each encoding is therefore swept over bond caps and reported against the compression ratio $\rho = \text{parameters}/4^L$ it achieves. Table II lists every run. The reported error is the relative $L_2$ difference from a dense reference integrated with the same scheme, step and velocity, averaged over the final third of the run, where it has saturated.

Alongside each evolution error is the error of compressing the *exact* field at the same budget, which separates the cost of representing the solution from the additional cost of evolving in the representation.

## IV. PARAMETERS REQUIRED AT FIXED ACCURACY

The quantity of practical interest is the budget required to reach a given accuracy. Table III reports the ratio $\rho_{\text{std}}/\rho_{\text{hybrid}}$, with the head size that achieves the smaller budget, obtained by log–log interpolation of $\rho(\text{error})$. Targets are listed only where both encodings span the target error.

The error ratio read at fixed $\rho$ instead of fixed accuracy is considerably larger — at $L = 10$ it runs from 2.8 at $\rho = 0.01$ to 25 at $\rho = 0.05$ — and it grows with resolution. Both quantities describe the same pair of curves. The rate–distortion curves are steep and steepen with $L$, so a horizontal offset of a factor 2 in $\rho$ appears as one to two decades along the error axis. The budget ratio is the quantity that answers how many parameters the head buys.

## V. EVOLUTION AGAINST REPRESENTATION

Figure 2 collects the comparison across resolutions. The lower middle panel plots the evolution error against the representation error at the same budget for every run. All points lie between the 1× and 25× lines, with most below 5×, across four resolutions, five head sizes and five bond caps.

$L = 8$ $(256^2)$

| $N$ | $\chi$ | $\rho$ | head % | $L_2$ (evolution) | $L_2$ (representation) |
|---|---|---|---|---|---|
| 0 | 8 | 0.0221 | – | $7.96 \times 10^{-1}$ | $1.62 \times 10^{-1}$ |
| 0 | 12 | 0.0436 | – | $3.35 \times 10^{-1}$ | $6.92 \times 10^{-2}$ |
| 0 | 16 | 0.0729 | – | $1.22 \times 10^{-1}$ | $2.93 \times 10^{-2}$ |
| 0 | 24 | 0.1393 | – | $2.64 \times 10^{-2}$ | $7.30 \times 10^{-3}$ |
| 0 | 32 | 0.2291 | – | $3.57 \times 10^{-3}$ | $1.41 \times 10^{-3}$ |
| 4 | 8 | 0.0208 | 9 | $7.79 \times 10^{-1}$ | $1.62 \times 10^{-1}$ |
| 4 | 12 | 0.0423 | 7 | $3.22 \times 10^{-1}$ | $6.92 \times 10^{-2}$ |
| 4 | 16 | 0.0716 | 5 | $1.24 \times 10^{-1}$ | $2.93 \times 10^{-2}$ |
| 4 | 24 | 0.1380 | 3 | $2.66 \times 10^{-2}$ | $7.30 \times 10^{-3}$ |
| 4 | 32 | 0.2278 | 2 | $3.57 \times 10^{-3}$ | $1.41 \times 10^{-3}$ |
| 6 | 8 | 0.0228 | 34 | $3.44 \times 10^{-1}$ | $6.79 \times 10^{-2}$ |
| 6 | 12 | 0.0423 | 28 | $1.65 \times 10^{-1}$ | $2.98 \times 10^{-2}$ |
| 6 | 16 | 0.0677 | 23 | $6.76 \times 10^{-2}$ | $1.54 \times 10^{-2}$ |
| 6 | 24 | 0.1282 | 18 | $1.38 \times 10^{-2}$ | $4.34 \times 10^{-3}$ |
| 6 | 32 | 0.2083 | 15 | $3.58 \times 10^{-3}$ | $1.41 \times 10^{-3}$ |
| 8 | 8 | 0.0423 | 74 | $6.40 \times 10^{-2}$ | $1.57 \times 10^{-2}$ |
| 8 | 12 | 0.0687 | 68 | $1.39 \times 10^{-2}$ | $4.38 \times 10^{-3}$ |
| 8 | 16 | 0.0989 | 63 | $3.63 \times 10^{-3}$ | $1.59 \times 10^{-3}$ |
| 8 | 24 | 0.1634 | 57 | $5.21 \times 10^{-4}$ | $2.73 \times 10^{-4}$ |
| 8 | 32 | 0.2396 | 52 | $9.85 \times 10^{-5}$ | $5.14 \times 10^{-5}$ |
| 10 | 4 | 0.0648 | 97 | $1.30 \times 10^{-1}$ | $2.15 \times 10^{-2}$ |
| 10 | 6 | 0.0981 | 96 | $1.09 \times 10^{-2}$ | $4.78 \times 10^{-3}$ |
| 10 | 8 | 0.1321 | 95 | $5.32 \times 10^{-3}$ | $2.08 \times 10^{-3}$ |
| 10 | 12 | 0.2005 | 94 | $6.41 \times 10^{-4}$ | $3.46 \times 10^{-4}$ |
| 10 | 16 | 0.2708 | 92 | $1.00 \times 10^{-4}$ | $6.87 \times 10^{-5}$ |

$L = 9$ $(512^2)$

| $N$ | $\chi$ | $\rho$ | head % | $L_2$ (evolution) | $L_2$ (representation) |
|---|---|---|---|---|---|
| 0 | 12 | 0.0131 | – | $7.23 \times 10^{-1}$ | $1.36 \times 10^{-1}$ |
| 0 | 16 | 0.0221 | – | $4.20 \times 10^{-1}$ | $7.39 \times 10^{-2}$ |
| 0 | 24 | 0.0436 | – | $1.50 \times 10^{-1}$ | $2.33 \times 10^{-2}$ |
| 0 | 32 | 0.0729 | – | $3.95 \times 10^{-2}$ | $6.65 \times 10^{-3}$ |
| 8 | 12 | 0.0194 | 61 | $1.35 \times 10^{-1}$ | $1.85 \times 10^{-2}$ |
| 8 | 16 | 0.0286 | 55 | $6.20 \times 10^{-2}$ | $9.17 \times 10^{-3}$ |
| 8 | 24 | 0.0496 | 47 | $1.30 \times 10^{-2}$ | $2.30 \times 10^{-3}$ |
| 8 | 32 | 0.0755 | 41 | $3.13 \times 10^{-3}$ | $7.48 \times 10^{-4}$ |
| 10 | 12 | 0.0523 | 90 | $1.30 \times 10^{-2}$ | $1.93 \times 10^{-3}$ |
| 10 | 16 | 0.0716 | 87 | $2.69 \times 10^{-3}$ | $7.28 \times 10^{-4}$ |
| 10 | 24 | 0.1112 | 84 | $3.47 \times 10^{-4}$ | $1.19 \times 10^{-4}$ |
| 10 | 32 | 0.1536 | 81 | $5.32 \times 10^{-5}$ | $1.98 \times 10^{-5}$ |

$L = 10$ $(1024^2)$

| $N$ | $\chi$ | $\rho$ | head % | $L_2$ (evolution) | $L_2$ (representation) |
|---|---|---|---|---|---|
| 0 | 16 | 0.0065 | – | $8.28 \times 10^{-1}$ | $1.15 \times 10^{-1}$ |
| 0 | 24 | 0.0131 | – | $3.96 \times 10^{-1}$ | $4.51 \times 10^{-2}$ |
| 0 | 32 | 0.0221 | – | $1.70 \times 10^{-1}$ | $1.73 \times 10^{-2}$ |
| 0 | 48 | 0.0436 | – | $3.54 \times 10^{-2}$ | $3.55 \times 10^{-3}$ |
| 0 | 64 | 0.0729 | – | $5.46 \times 10^{-3}$ | $6.41 \times 10^{-4}$ |
| 8 | 16 | 0.0081 | 48 | $2.62 \times 10^{-1}$ | $2.54 \times 10^{-2}$ |
| 8 | 24 | 0.0146 | 40 | $1.03 \times 10^{-1}$ | $9.67 \times 10^{-3}$ |
| 8 | 32 | 0.0228 | 34 | $4.41 \times 10^{-2}$ | $4.10 \times 10^{-3}$ |
| 8 | 48 | 0.0423 | 28 | $7.60 \times 10^{-3}$ | $8.50 \times 10^{-4}$ |
| 8 | 64 | 0.0677 | 23 | $1.56 \times 10^{-3}$ | $2.16 \times 10^{-4}$ |
| 10 | 16 | 0.0189 | 83 | $5.19 \times 10^{-2}$ | $3.77 \times 10^{-3}$ |
| 10 | 24 | 0.0300 | 78 | $1.02 \times 10^{-2}$ | $9.06 \times 10^{-4}$ |
| 10 | 32 | 0.0423 | 74 | $2.03 \times 10^{-3}$ | $2.66 \times 10^{-4}$ |
| 10 | 48 | 0.0687 | 68 | $1.62 \times 10^{-4}$ | $3.19 \times 10^{-5}$ |
| 10 | 64 | 0.0990 | 63 | $1.92 \times 10^{-5}$ | $4.68 \times 10^{-6}$ |
| 12 | 16 | 0.0648 | 96 | $2.27 \times 10^{-3}$ | $3.20 \times 10^{-4}$ |
| 12 | 24 | 0.0981 | 96 | $1.80 \times 10^{-4}$ | $3.71 \times 10^{-5}$ |
| 12 | 32 | 0.1322 | 95 | $2.26 \times 10^{-5}$ | $6.34 \times 10^{-6}$ |
| 12 | 48 | 0.2005 | 94 | $6.80 \times 10^{-7}$ | $2.46 \times 10^{-7}$ |
| 12 | 64 | 0.2708 | 92 | $3.26 \times 10^{-8}$ | $1.06 \times 10^{-8}$ |

$L = 11$ $(2048^2)$

| $N$ | $\chi$ | $\rho$ | head % | $L_2$ (evolution) | $L_2$ (representation) |
|---|---|---|---|---|---|

| target $L_2$ | $L = 8$ | $L = 9$ | $L = 10$ | $L = 11$ |
|---|---|---|---|---|
| $1 \times 10^{-1}$ | 1.4 ($N$=6) | 2.3 ($N$=8) | 1.9 ($N$=8) | – |
| $6 \times 10^{-2}$ | 2.3 ($N$=8) | 2.1 ($N$=8) | 1.8 ($N$=8) | 2.1 ($N$=10) |
| $3 \times 10^{-2}$ | 2.5 ($N$=8) | – | 2.1 ($N$=10) | – |
| $1 \times 10^{-2}$ | 2.4 ($N$=8) | – | 2.1 ($N$=10) | – |
| $3 \times 10^{-3}$ | – | – | – | – |

TABLE III. Parameters saved at fixed accuracy, and the head size achieving it. The saving is 1.8–2.5 and shows no trend with resolution over $256^2$–$2048^2$.

The practical consequence is that the ordering of encodings in $N$ and the scaling in $L$ can be established by compressing stored fields, which costs minutes, rather than by integrating them, which costs core-days. The representation measurement does not substitute for the evolution measurement when the absolute error level matters.

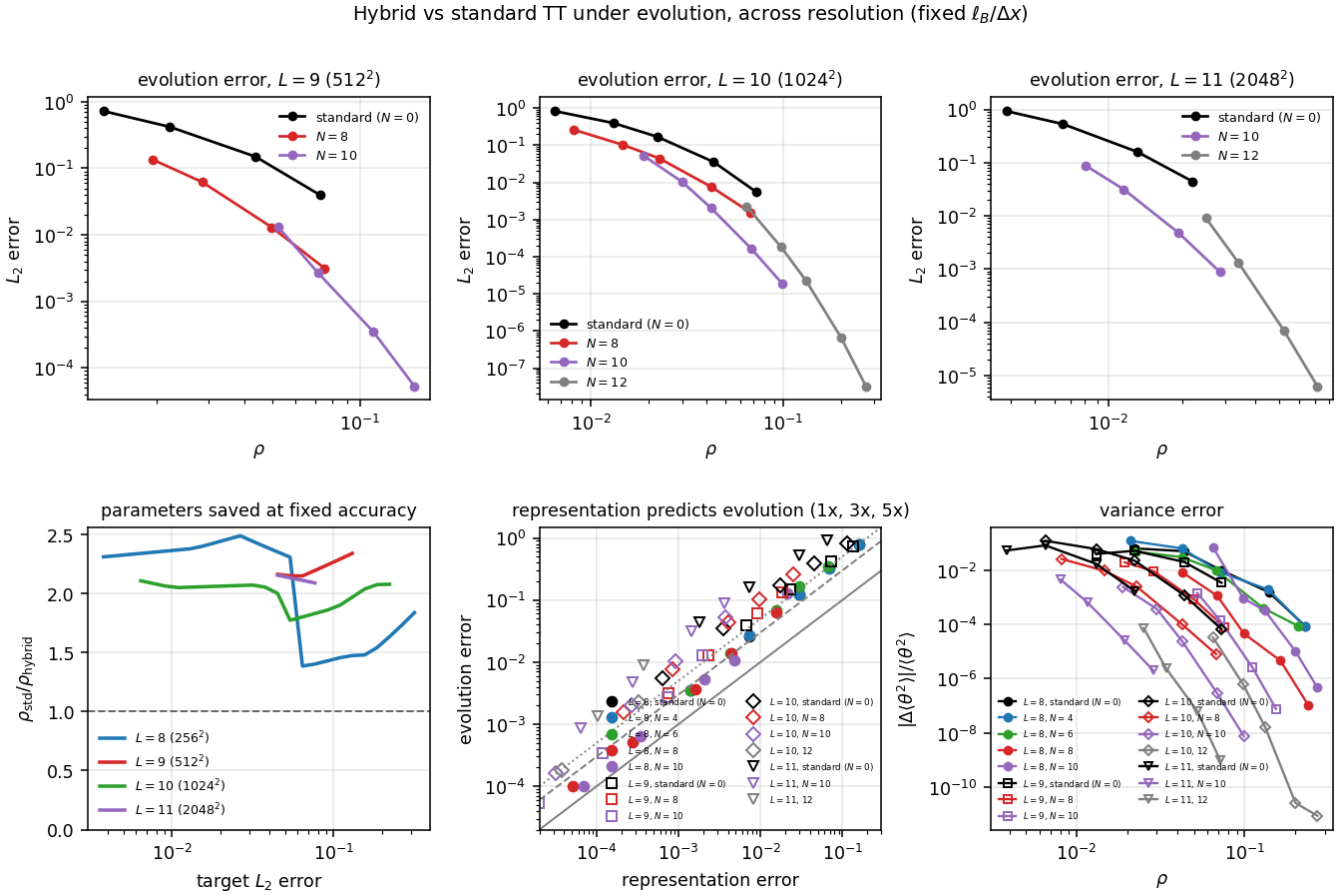



FIG. 2. Top: rate–distortion at $L = 9, 10, 11$. Bottom left: parameters saved at fixed accuracy. Bottom centre: evolution error against representation error at the same budget, with 1×, 3× and 5× guides. Bottom right: relative error in $\langle\theta^2\rangle$.

## VI. COST

Table IV gives measured cost per step at $\chi = 32$. At equal bond cap the hybrid encoding is slightly cheaper than the standard one, since it carries $2L - N$ tail cores instead of $2L$, while the head contributes $O(2^N r^2 \chi)$ through the un-fusing path. At equal $\rho$ the comparison is closer, because reaching a given $\rho$ with a larger head permits a smaller $\chi$.

All runs are single-threaded. On a 256-core node, four threads instead of one measured 29× slower per step, because

| $L$ | $N$ | $\chi$ | $\rho$ | ms/step |
|---|---|---|---|---|
| 8 | 0 | 32 | 0.2291 | 846 |
| 8 | 4 | 32 | 0.2278 | 920 |
| 8 | 6 | 32 | 0.2083 | 822 |
| 8 | 8 | 32 | 0.2396 | 535 |
| 9 | 0 | 32 | 0.0729 | 1507 |
| 9 | 8 | 32 | 0.0755 | 1419 |
| 9 | 10 | 32 | 0.1536 | 1518 |
| 10 | 0 | 32 | 0.0221 | 818 |
| 10 | 8 | 32 | 0.0228 | 940 |
| 10 | 10 | 32 | 0.0423 | 1068 |
| 10 | 12 | 32 | 0.1322 | 1989 |
| 11 | 0 | 32 | 0.0065 | 2257 |
| 11 | 10 | 32 | 0.0116 | 3348 |
| 11 | 12 | 32 | 0.0340 | 3596 |

TABLE IV. Measured cost per step at $\chi = 32$. Runs shared cluster nodes, so absolute values carry scheduling noise; the comparison within a row of constant $L$ is the meaningful one.

a step consists of thousands of small contractions and SVDs, none large enough to amortise thread dispatch.

## VII. SCOPE OF THESE RESULTS

The measurements above are specific to the flow used. The cellular flow $\psi = \sin x \sin y$ has closed streamlines and is integrable, and its scalar spectrum is steep, $E_\theta(k) \sim k^{-4.9}$ over $3 \le k \le 20$, rather than the $k^{-1}$ Batchelor form of a chaotic flow. Breaking the streamlines with $\psi = \sin(x + B \sin \omega t) \sin y$ raises the rank plateau from 75 to 84 at $L = 8$ and leaves the spectrum steep.

The size of the hybrid advantage depends on the field. On random-phase fields of prescribed spectral slope the same measurement gives 1.2 at $k^{-5/3}$ and 3.5 at $k^{-5}$, against 5–24 at fixed $\rho$ for the cellular field. The difference is attributable to spatial coherence rather than to spectral slope or to alignment with the dyadic block boundaries: rigid shifts of the field by 2, 4, 8 and 13 cells leave the advantage unchanged.

Real turbulent scalar fields lie between the two cases. The representation measurements on DNS data reported in the paper remain the reference point for what to expect there; the contribution of the present benchmark is that the advantage measured for representation is preserved under evolution.

## VIII. CONFIGURATION

- Grids $2^L \times 2^L$ with $L = 8, 9, 10, 11$; $\kappa = 5 \times 10^{-3}$, $1.25 \times 10^{-3}$, $3.125 \times 10^{-4}$, $7.8125 \times 10^{-5}$.
- $\Delta t = 0.2\,\Delta x$ (CFL-limited at every resolution), $T = 8$, giving 1630, 3259, 6519 and 13038 steps.
- Head sizes $N \in \{0, 4, 6, 8, 10, 12\}$, bond caps $\chi \in \{4, \ldots, 64\}$ applied uniformly as `bd_first` = `bd_remaining`, with `renormalize` = `False` so that truncated norm is not restored.
- 69 runs in total; the $L = 10$ and $L = 11$ sets were run as Slurm arrays, one core per task.